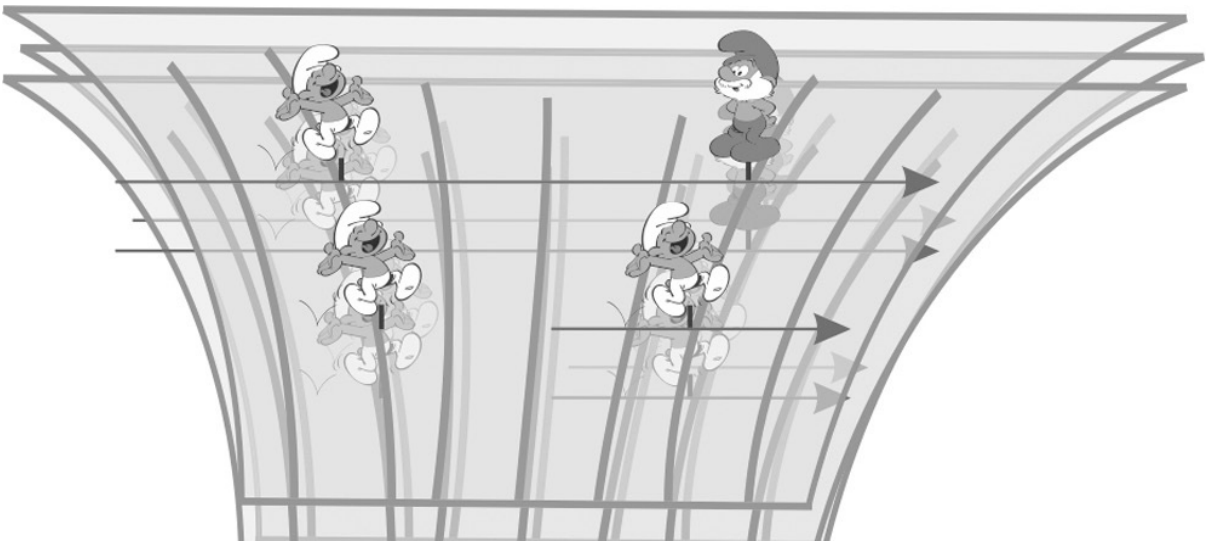
quanta of space and time. Then I spent the rest of my life actually trying to achieve this.

QUANTUM SUPERPOSITIONS OF TIMES

The second discovery made by quantum mechanics is *indeterminacy*: it is not possible to predict exactly, for instance, where an electron will appear tomorrow. Between one appearance and another, the electron has no precise position,[55] as if it were dispersed in a cloud of probability. In the jargon of physicists, we say that it is in a "superposition" of positions.

Spacetime is a physical object like an electron. It, too, fluctuates. It, too, can be in a "superposition" of different configurations. The illustration of stretched time at the end of chapter 4, for example—if we take quantum mechanics into account—should be imagined as a blurred superposition of different spacetimes, more or less as shown below.



Similarly, the structure of light cones fluctuates at all points that distinguish among past, present, and future.